

Assignment-0

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1 Setup

The experiments were conducted on the following hardware and software configuration.

Table 1: Experimental Setup

Component	Specification
CPU Model	<i>12th Gen Intel(R) Core(TM) i7 - 12700H</i>
CPU Cores / Threads	<i>14 cores, 20 threads</i>
RAM	<i>15.6 GB</i>
Storage	<i>477 GB</i>
Operating System	<i>Windows 11</i>
PostgreSQL Version	PostgreSQL 18.1
Architecture	x86_64, 64-bit

2 Database

Name: Join Order Benchmark (2013 Snapshot)

The experiments use the IMDb database from the 2013 snapshot of the Join Order Benchmark (JOB). The database contains 21 tables. The cardinality of each table was obtained by executing a `COUNT(*)` query on the corresponding table.

Table 2: JOB Database Tables, Primary Keys, and Cardinalities

Table Name	Primary Key	Cardinality
aka_name	id	901,343
aka_title	id	361,472
cast_info	id	36,244,344
char_name	id	3,140,339
comp_cast_type	id	4
company_name	id	234,997
company_type	id	4
complete_cast	id	135,086
info_type	id	113
keyword	id	134,170
kind_type	id	7
link_type	id	18
movie_companies	id	2,609,129
movie_info	id	14,835,720
movie_info_idx	id	1,380,035
movie_keyword	id	4,523,930
movie_link	id	29,997
name	id	4,167,491
person_info	id	2,963,664
role_type	id	12
title	id	2,528,312